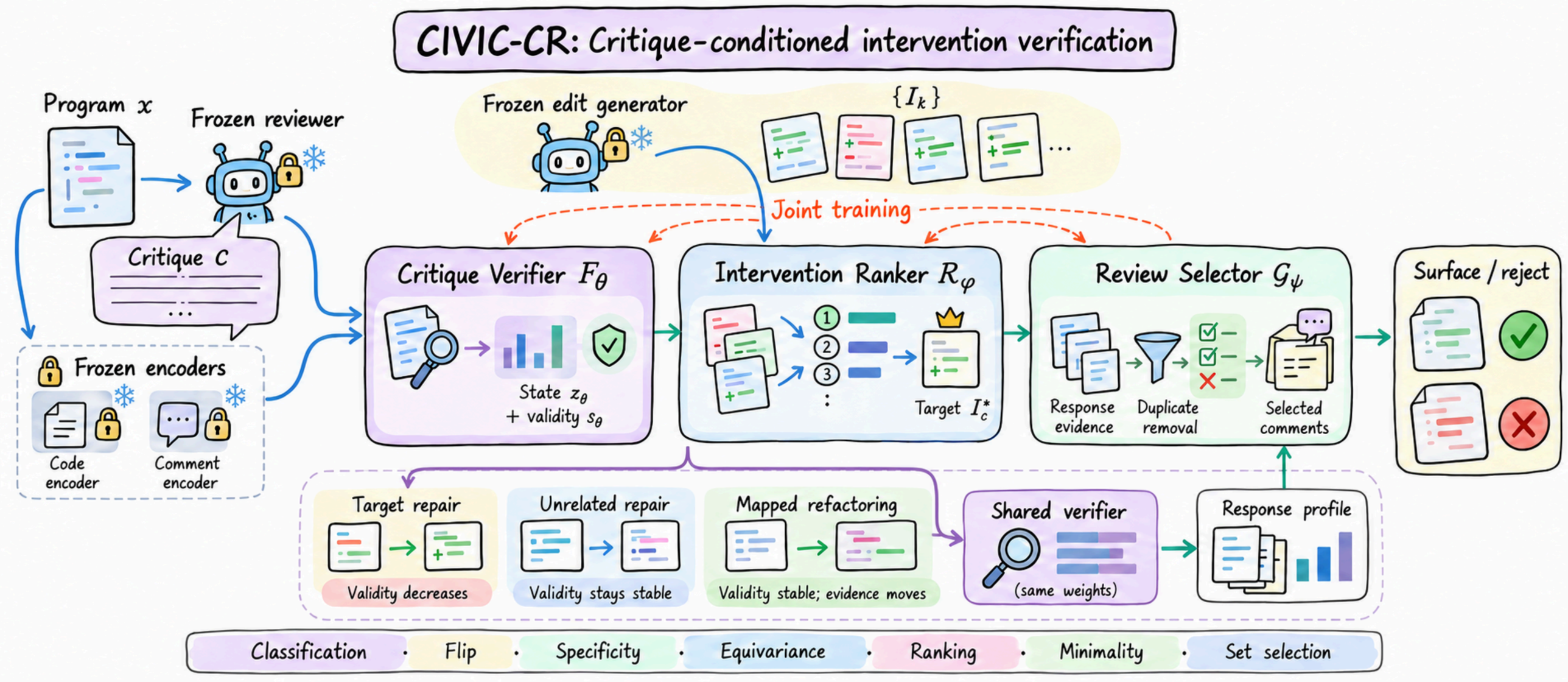

CIVIC-CR: Critique-conditioned intervention verification
Program x
Frozen reviewer
Critique C
...
Frozen encoders
Code encoder
Comment encoder
Frozen edit generator
$\{I_k\}$
Joint training
Critique Verifier F_θ
State z_θ
+ validity s_θ
Intervention Ranker R_φ
1
2
3
Target I_c^*
Review Selector G_ψ
Response evidence
Duplicate removal
Selected comments
Surface / reject
Target repair
Validity decreases
Unrelated repair
Validity stays stable
Mapped refactoring
Validity stable; evidence moves
Shared verifier
(same weights)
Response profile
Classification · Flip · Specificity · Equivariance · Ranking · Minimality · Set selection